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// enable interrupts before leaving
void incangle()
// increase delay by 1 ms
{
    EX1=0;
// all other things remains same as
above
    writemcmd(0x00);
    led4=1;
    if(d1<9)
    {
        d1++;
        d2+=18;
        writestr("angle:");
        display(d2);
        writestr("\n deg. ");
    }
    else if(d1==9) writestr("Max
phase angle ");
    led4=0;
    EX1=1;
}
void delay(int d)
// generates delay from 1 to 9 ms
{
    int k;
    TLO = 0x17;
// load timer 0 with 64535 = FCI17h
    TH0 = 0xFC;
// so it will overflow after 1000 counts
    TR0 = 1;
// start timer
    for(k=0;k<d;k++) // count
overflows of timer 0 till
    {
// desire delay is required
        while(TF0==0);
        TF0 = 0;
        TLO = 0x17;
        TH0 = 0xFC;
    }
    TR0 = 0;
}

void writemcmd(unsigned char a)
// send command to LCD
{
    busy();
// check busy flag
    rs = 0;
// select command register
    rw = 0;
// write enable
    P0 = a;
// send byte to LCD
    en = 1;
// apply strobe pulse
    en = 0;
}

void writedata(unsigned char b)
// send data to LCD
{
    busy();
// check busy flag
    rs = 1;
// select data register
    rw = 0;
// write enable
    P0 = b;
// send byte to LCD
    en = 1;
// apply strobe pulse
    en = 0;
}

void busy()
// check busy flag of LCD
{
    en = 0;
// disable display
    P0 = 0xFF;
// P0 as input
    rs = 0;
// select command register
    rw = 1;
// read enable
    while(b==1)
// if busy bit is 1
    {
        en=0;
// remain within loop
        en=1;
    }
    en=0;
}

void writestr(unsigned char *s)
// send string message to LCD
{
    unsigned char l,i;
    l = strlen(s);
// get length of string
    for(i=0;i<l;i++)
    {
        writedata(*s);
// till the length of string
        s++;
// send characters one by one
    }
}

void display(unsigned int z)
// convert decimal number to
ASCII
{
    int z1,s,ASCII[3];
    if(z==100)
// if number is greater then 100
    {
        // take 3 digits
        a=2;
        while(z>=10)
        {
            z1=z%10;
            ASCII[a]=z1+0x30; // convert all three
into ASCII
            z=z/10;
            a--;
        }
        ASCII[0]=z+0x30;
    }
    else
// otherwise take 2 digits
    {
        z1=z%10;
        ASCII[2]=z1+0x30;
        z=z/10;
        ASCII[1]=z+0x30;
        ASCII[0]=z+0x30;
    }
// take first digit as 0
    writedata(ASCII[0]);
// display them one by one
    writedata(ASCII[1]);
    writedata(ASCII[2]);
}

void int0(void) interrupt 0
// external interrupt 0 subroutine
{
    switch(P1)
    {
        case 0xFE:
// for first key
        keyldy();
// after key debounce delay
        c++;
// increase counter
    }
    if((c%2)==1) // check even or odd
    {
        writemcmd(0x01); // for odd times
    }
}

writestr("SCR ON "); // switch on SCR
writemcmd(0x00);
writestr("angle:"); // display current
angle
display(d2);
writestr("\n deg. ");
led2=0; // make led2 off
EX1 = 1;
// enable external interrupt 1
}
else
{
    writemcmd(0x01); // for even times
    writestr("SCR OFF "); // SCR is off
    led2=1; // led2 is on
    EX1 = 0;
// disable external interrupt 1
}
break;
// for second key
keyldy();
// after key debounce
incangle();
// increase phase angle
break;
// for third key
keyldy();
// after key debounce
decangle();
// decrease phase angle
break;
}

void int1(void) interrupt 2
// external interrupt 1 subroutine
{
    int t;
    led1=1;
// led1 is on
    delay(d1);
// after desired delay
    pulse=1;
// send pulse on p2.0
    for(t=0;t<200;t++);
    pulse=0;
    led1 = 0;
// led1 off
}

void main()
{
    TMOD = 0x01;
// initialize timer0 as 16 bit timer
    P2=0x00;
// P2 and P0 as output ports
    P0=0x00;
    P1=0xFF; // P1 as input port
    writemcmd(0x3C);
// initialize LCD
    writemcmd(0x0E);
    writemcmd(0x01);
    writestr("SCR Phase angle");
// display message at the center
    writemcmd(0x04);
    writestr("Control");
// of LCD
    led2=1;
// scr off led is on
    IE=0x81;
// enable external interrupt 0
    while(1);
// continuous loop
}

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